



University of Sri Jayewardenepura

Faculty of Technology

Department of Information Communication Technology

ITS 4243 - Microservices and Cloud Computing

Assignment 02

Title: Smart Travel Booking Platform – Inter-Service Communication Using REST API, Feign Client, and WebClient

Objective:

Build a distributed travel booking backend platform containing 6 microservices.

Microservices (6):

1. User Service
2. Flight Service
3. Hotel Service
4. Payment Service
5. Notification Service
6. Booking Service (Main Orchestrator)

Communication Rules:

- No deprecated technologies (no RestTemplate)
- REST calls allowed only via WebClient and standard controllers
- Feign Client must be used for at least two service interactions

Communication Flow:

1. Booking Service → User Service (WebClient)

2. Booking Service → Flight Service (Feign Client)
3. Booking Service → Hotel Service (Feign Client)
4. Booking Service → Notification Service (WebClient REST call)
5. Payment Service → Booking Service (WebClient)

Assignment Tasks:

- Implement all 6 microservices in Spring Boot 3+ with Java 17+
- Use Feign Client for Flight and Hotel communication
- Use WebClient for:
 - Booking → User communication
 - Booking → Notification communication
 - Payment → Booking communication
- Implement Controllers, Services, DTOs, Entities, and Exception Handling
- Use proper response models and error responses
- Add Swagger (optional)
- Submit Postman Collection
- Submit README with architecture diagram
- Each service must run on its own port

Booking Flow Summary:

1. User sends a booking request:

```
{  
  "userId": 1,  
  "flightId": 200,  
  "hotelId": 55,  
  "travelDate": "2025-01-10"  
}
```

2. Booking Service:

- Validates user via WebClient
- Checks flight availability via Feign Client
- Checks hotel availability via Feign Client

- Calculates total cost
- Stores booking as PENDING
- Calls Payment Service via WebClient
- Calls Notification Service via WebClient
- Updates booking to CONFIRMED

Folder Structure:

smart-travel-platform/

└─ user-service/

└─ flight-service/

└─ hotel-service/

└─ booking-service/

└─ payment-service/

└─ notification-service/

Submission Requirements:

- Final GitHub repository
- Postman collection
- PDF/README explaining communication flow
- Screenshots of successful interactions

Deadline : Please submit on or before 13th of December 2025.